Generation Probabilities are Not Enough: Improving Error Highlighting in AI Code Suggestions [1]

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Programmers increasingly use AI generated code (e.g., from CoPilot) in IDEs! [7]

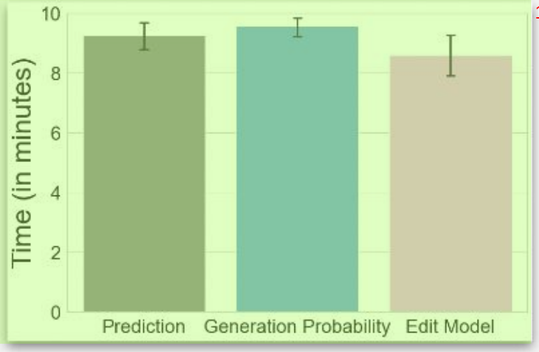
However, generated code may be erroneous! ***We need to help programmers double-check AI code and identify when and where it may be incorrect.*** [8]

We explore two options to highlight uncertain, potentially erroneous regions: [9]

- *Use* **generation probabilities** *or "confidence scores" to show where the model is uncertain* [10]
- Learn an **edit model** to show where programmers are most likely to make edits

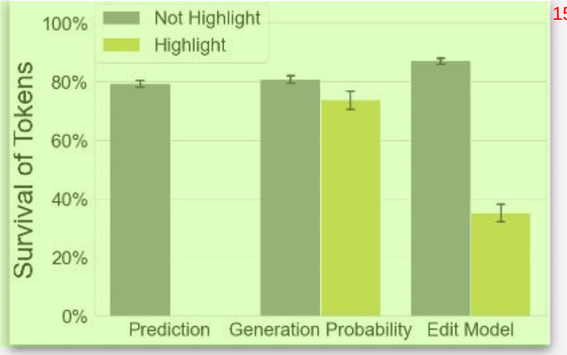
Mixed-methods, within-subjects study with 30 programmers showed that ***edit model leads to***: [11]

- Significantly ***faster*** task completion time [12]



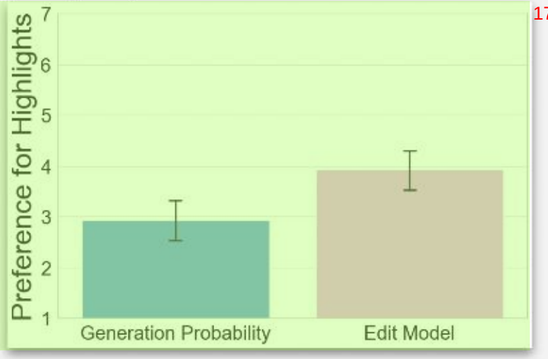
[13]

- Significantly ***more localized edits*** [14]



[15]

- Stronger ***preference*** [16]



[17]

compared to the generation probability and no-highlights [18]